

Exercise Sheet 5

Rasterization

Question 1.

With this exercise sheet, a framework is available for download in which programming tasks are to be solved. The framework renders a cube with a color gradient on the graphics card. The camera can be rotated and zoomed around the cube as usual.

(a) (20 points) The goal of this exercise is to write a software rasterizer that draws the same cube on the CPU. Pressing the 'S' key can switch between the two representations. The software representation should initially show a test image, which must be deleted (set `bool drawTestImage = false;` (l.119) for the final output. The necessary steps are:

- Transforming the vertices of each triangle into 'Normalized Device Coordinates' to determine the depth value.
- Transforming the vertices into screen coordinates.
- Using the scanline algorithm to rasterize triangles.
 - Additionally, interpolate the color and z-values for each set pixel.
 - Before setting a color value: perform a depth test (the method `getDepthValue` of the `software_framebuffer` object provides the current z-value for a pixel).
- Write the color value of a pixel to the 'back buffer' (can be done using the provided function `setColorValue`).
- Write the z-value of a pixel to the 'depth buffer' (can be done using the provided function `setDepthValue`).

To simplify, the scene was created so that some aspects do not need to be considered:

- Clipping: All triangles should ALWAYS be displayed within the screen.
- Backface Culling: The back faces of the cube will always be overwritten by front faces due to the depth test.

All areas where you need to change the source code are marked with a comment in the form `// TODO:.`

Submit your solution by **Monday, 14.07.2025, 23:55** via the Moodle platform in the form of a ZIP file.